

Basic Electrical and Electronics Engineering

SITAMS CSE (AI & ML) • R23 • I Year • II Semester • Course Code: 23ESC112T

Source basis: Official SITAMS R23 CSE (AI & ML) syllabus uploaded for this Study Hub. This concise revision guide preserves the syllabus topic statements and adds a study checklist; it is not a replacement for prescribed textbooks or faculty notes.

Unit Part A • Unit 1: DC and AC Circuits

Syllabus topics

- Electrical circuit elements
- Ohm's law
- KCL and KVL
- series, parallel and series-parallel circuits
- superposition
- AC fundamentals
- RMS and average values
- phasors
- impedance
- active, reactive and apparent power
- power factor

Revision checklist

- Know the definitions and principles named in the syllabus.
- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

Unit Part A • Unit 2: Machines and Measuring Instruments

Syllabus topics

- DC motor, DC generator, single-phase transformer, three-phase induction motor and alternator
- moving-coil and moving-iron instruments
- wattmeter, energy meter and Wheatstone bridge

Revision checklist

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Unit Part A • Unit 3: Energy Resources, Electricity Bill and Safety Measures

Syllabus topics

- Conventional and non-conventional energy resources

- hydel, thermal, nuclear, solar and wind generation
- household power ratings
- electricity units and tariff
- domestic billing
- fuse, MCB, electric shock, earthing and safety precautions

Revision checklist

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Unit Part B • Unit 1: Semiconductor Devices

Syllabus topics

- Evolution of electronics
- PN junction diode and Zener diode
- transistor configurations and characteristics
- elementary small-signal CE amplifier

Revision checklist

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- Prepare the standard methods, diagrams, formulas or algorithms relevant to the listed topics.
- Practice representative examples or problems from the prescribed study material.
- Review this unit against the official SITAMS syllabus before attempting its quiz.

Unit Part B • Unit 2: Basic Electronic Circuits and Instrumentation

Syllabus topics

- DC power supply and bridge rectifier
- capacitor filter
- Zener regulator
- public-address system
- common-emitter amplifier and frequency response
- electronic instrumentation system

Revision checklist

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Unit Part B • Unit 3: Digital Electronics

Syllabus topics

- Number systems

- logic and universal gates
- BCD, Excess-3, Gray and Hamming codes
- Boolean algebra
- half/full adders
- sequential circuits
- flip-flops, registers and counters

Revision checklist

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Study Hub status

This guide is syllabus-derived. External resources and quiz questions are tracked separately and are only labelled verified after source checking.